Supplementary Material for AlgoTutorGen: Contract-Guided Compositional Synthesis of Verifiable Interactive Algorithm Tutors

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Abstract

This supplement documents protocols, secondary results, artifact provenance, and reproducibility boundaries; it adds no primary claim. Its finite executable audits do not establish universal guarantees, LOCAL RESUME is not fully checkpointed recovery, and all human studies remain incomplete.

Supplement Scope and Reading Guide

The main paper contains every result required to evaluate the contribution within the seven-page technical-content limit. This document expands the measurement protocol, reports secondary tables, records artifact provenance, and presents frozen English case studies. Review of this supplement is not necessary to interpret the primary claims. Where an audit set contains a targeted retry, we identify it explicitly rather than converting it into an original generation success.

Machine OK Protocol and Artifact Provenance

Decision Rule

For each page, the browser harness records nine booleans: page load, visible-answer match, reachable interaction, correct feedback, wrong feedback, hint behavior, answer-reveal behavior, learning-log behavior, and protected-answer stability. A page is MACHINE OK only if all nine are true. Correct and wrong feedback are tested using distinct submitted values. Hint and answer-reveal controls must change their associated pedagogical state, and the learning log must record the tested event. The ninth release check compares the protected visible-answer summary before and after the audited teaching actions. It is deliberately lighter than the independent noninterference stress suite, which compares algorithm-state, current-step, and artifact hashes after each action in 100 random sequences per page. Navigation in that stronger suite may change the step only to a verified target frame. External requests are blocked during both audits.

Generation and Statistical Boundaries

The main comparison uses sample index 0 for each of 200 tasks, preventing tasks with more stored inputs from receiving greater weight. The full benchmark contains 646 samples. The original ALGOTUTORGEN condition uses DeepSeek-V4-Pro. Cross-model final-quality runs use DeepSeek-V4-Flash, GLM-5.2, and Kimi-K2.5: each starts from two candidates and two repair rounds, and failed tasks may receive a recorded retry with at most three candidates and three rounds. Consequently, the final cross-model table is not a uniform-call or token-matched comparison. Direct HTML receives the concrete input and expected answer and is explicitly asked to generate the complete interaction suite. The main selected-final ALGOTUTORGEN set contains 195 primary successes plus five recorded targeted retries; it must not be described as 200 first-pass successes.

McNemar tests use paired task-level binary outcomes. Paired bootstrap confidence intervals resample tasks, not frames, actions, or individual metric checks. Holm correction applies to test families, and paired ordinal or continuous secondary scores use Wilcoxon signed-rank tests where appropriate. Frame and action totals describe implementation coverage and are not treated as independent inferential samples. A nonsignificant result is not evidence of equivalence.

Executable Stage Interfaces and Configuration

Table S1 states the implementation boundary of every stage. The optional correctness-contract drafter is available in the codebase, but the main Full-200 experiment is not described as uniformly passing through it. The principal path begins with the executable solution/trace specification.

JSON generation uses temperature 0.2 and text generation uses 0.3, except Kimi models, for which the provider-required value is 1.0. The API client allows four JSON-parse retries and two transport retries by default; these calls are included in logged model-call metadata when present. Remote model versions can drift.

Sandbox and Threat Model

Generated `solve`, `trace`, and optional verifier functions are parsed with an AST check, executed in a separate Python child process, and receive a restricted builtin namespace plus allowlisted standard-library modules. Each function has a

Table S1: Executable stage interfaces. “Generated” denotes an open-ended model output; “deterministic” denotes project code operating on validated data.

Stage	Input	Output	Construction	Blocking boundary and failure handling
Solution/trace generation	Problem, concrete input, optional expected result, requested variants	`solve` plus `trace` JSON	Generated	Strict schema and function signatures; malformed JSON is retried, then candidate-level generation/repair proceeds
Materialization	Solution specification	Typed SemanticTrace and solver result	Sandboxed execution	Result agreement, trace schema/references, process continuity, demo readiness, and multi-variant consistency
Scene compilation	Validated SemanticTrace	SceneGraph frames	Deterministic	One event maps to one frame; structural scene gate checks nonempty/visible objects and references
Teaching enrichment	Selected digest of compiled, verified frames	Teaching/interaction overlay applied to the scene	Generated	Strict overlay schema and sanitization; primary 30-frame context retries with three frames; failure remains a warning
Creative Stage2	Artifact/frame context	Scoped stage CSS/template/renderer	Generated	Prompt restricts output to presentation; the sanitizer checks required scripts and stable IDs, but the supplied context is not a deep-frozen security boundary
Export and runtime	Artifact, scene, sanitized teaching	Self-contained HTML and sidecar JSON	Deterministic	Release gate, external-resource policy, and browser audit

30-second execution timeout, and a trace event whose serialized state exceeds 20,000 characters is rejected. The DSL guard also rejects undocumented object methods before trace execution. This boundary is intended to contain accidental model-generated code and enforce the artifact contract; it is not an operating-system or container security claim against a determined adversary.

Prompt Engineering

Table S3 follows the module-first presentation used in the reference paper: it exposes each prompt’s context, output, immutable boundary, and retry trigger before showing representative instructions. Token-exact production sources are the listed files. Some production prompts are Chinese; the English text below and in `artifacts/paper_case_studies_en/PROMPT_TEMPLATES_EN.md` is a faithful structural rendering, not a byte-identical transcript.

Representative English Prompt Skeletons

Executable specification.

```
ROLE: Generate executable solve(input_data) and
trace(input_data).
RETURN: one JSON object with problem_title,
input_contract, verifier_code, variants.
TRACE: use only documented TraceSession DSL
methods; return sess.to_trace().
CHECK: solve/trace answers agree; events point to
real solve code lines; JSON is complete.
FORBID: HTML, CSS, JavaScript, rendering fields,
undocumented methods, or extra keys.
```

Teaching overlay.

```
SOURCE OF TRUTH: the validated SemanticTrace
digest.
RETURN: frames[step, teaching, interaction] only.
TEACHING: what, why, formula, invariant,
common_mistake, hint.
INTERACTION: at most one grounded prediction per
selected frame.
FORBID: changing op, targets, deps, state, result,
or code_line.
```

Creative Stage2.

```
DRAW: only the problem-specific main stage inside
the fixed shell.
READ: ctx.frame, ctx.state, ctx.evidence,
ctx.input, ctx.result.
RETURN: scoped style, optional template,
renderCreativeStage(ctx).
PROTOCOL: return no full page or duplicate
controls; do not use network/storage, recompute
answers, or intentionally mutate facts.
ENFORCEMENT: sanitizer checks required scripts
and stable IDs; nested context values are not
deep-frozen.
```

Direct HTML baseline.

```
RETURN: one complete offline HTML tutor.
INCLUDE: code, current state, timeline,
navigation, checkpoint, distinct feedback, hint,
show answer, learning log, and final answer.
INPUT: the same concrete task, input, expected
result, and teaching requirements.
REPAIR: replace the whole HTML document after
structured failure; no trace/scene checkpoint is
available.
```

Table S2: Recorded generation and execution configuration. Counts are maximum policy budgets, not a claim that every task consumed the maximum.

Condition	Model	Candidate/repair policy	Runtime limits	Important boundary
AlgoTutorGen main	DeepSeek-V4-Pro	2 solution variants; 2 candidates; 2 repair rounds per candidate	32,768 output tokens; 900 s API timeout; 3,000 s task timeout; concurrency 8	Strict warnings; teaching enabled; sample index 0
Direct HTML main	DeepSeek-V4-Pro	1 candidate; at most 2 repairs	32,768 output tokens; 2,400 s task timeout; concurrency 8	Expected result visible; no browser-feedback repair; self-contained HTML requested
Cross-model AlgoTutorGen	Flash, GLM, Kimi	Initial 2×2; failed tasks eligible for final-quality retry up to 3×3	Browser audit sharded 16 ways	Final frozen pages reported; calls/tokens differ from Direct
Held-out AlgoTutorGen	DeepSeek-V4-Pro	2 candidates; 2 repair rounds	900 s API timeout; 2,400 s task timeout; concurrency 8	Forty new tasks from 15 families
WebGen-Agent	DeepSeek-V4-Pro	At most 5 agent iterations	Same final browser audit	Task-adapted external code path; not a reproduction of its original benchmark
HTMLCure strict	DeepSeek repair; Gemini evaluator	1 repair iteration	External requests blocked in final audit	Operates on frozen Direct pages; self-containment is part of strict failure
Direct-BrowserRepair	DeepSeek-V4-Pro	Separate 1/2/3/5-call budgets	12,000 repair tokens; 80k target per task	The 1-call condition has not yet consumed browser feedback

Table S3: Prompt modules and hard boundaries.

Module	Context	Required output	Hard boundary	Retry trigger
Executable spec	Problem, input, optional expected, requested variants	Strict solution/trace JSON	Fixed fields; exact signatures; documented `TraceSession` DSL; no HTML/rendering	JSON parse, schema, sandbox, result, process, or release failure
Optional correctness contract	Problem, input, optional expected	`correctness-contract-v1` JSON	Expected has priority; oracle returns answer and is not a proof	Schema/oracle failure
Spec repair	Previous complete JSON plus structured diagnostics	Complete replacement solution/trace JSON	Minimal targeted change; preserve signatures and DSL; no invented methods	Bounded round while candidate remains
Teaching overlay	Digest of verified frames	`step`, `teaching`, `interaction` only	Read-only algorithm facts; checkpoint answer must follow from supplied frame	Parse/schema failure; retry with three frames
Creative Stage2	Verified artifact and current frame context	Scoped style/template/render function	Prompt-level presentation-only constraint; sanitizer checks required scripts and stable IDs	Layout overlap/clipping report
Direct HTML baseline	Same task/input/expected and teaching requirements	Complete self-contained HTML	No private contract claim; all page behavior generated together	Whole-page structured repair
Screenshot judge	Screenshot and rubric	Strict score JSON	Visible evidence only; no correctness or release decision	Parse retry only

Nested Contract Survival

The nested levels are: C1 visible-answer match; C2 C1 plus page load; C3 C2 plus reachable interaction; C4 C3 plus correct and wrong feedback; C5 C4 plus hint, answer reveal, and learning log; and C6 C5 plus protected-answer stability. Values in Table S5 are cumulative. Conditional survival at level i is the cumulative Ci count divided by the cumulative C$(i-1)$ count; no independence assumption is made. C6 is not the stronger full-state noninterference stress test.

The repeated pattern is not that all contracts are independent. Rather, monolithic pages continue to lose surviving artifacts at load, interaction, and feedback boundaries, while factored pages that pass semantic validation retain most downstream mass.

Theory-Aligned Executable Audits

Representation Preservation and Determinism

The preservation audit compares the normalized algorithm-state projection stored in each trace event, scene frame, and reachable runtime frame. The projection includes stable identifiers, values, marks, pointers, graph edges, container contents, current step, and final answer; layout, CSS, and teaching text are excluded.

Table S4: Artifact sets used by the main and theory-aligned evaluations. The completed held-out audit set contains one documented targeted retry that is not counted as an original generation pass.

Set	Artifacts/pages	Frames/sequences	Primary use	Provenance boundary
Main benchmark	200	9,421 frames	Reliability, preservation, mutation	One sample per task across 23 families; selected-final artifacts
Held-out generation	39 valid of 40	–	Frozen-task generalization	Original result remains 39/40
Completed held-out audit	40	4,568 frames	Semantic/state audits	Fortieth artifact is a documented targeted retry
Long-trace	54	41,119 frames	Preservation and scalability	18 tasks at small, medium, and large scales
Noninterference	240 pages	24,000 sequences	Browser state isolation	Main 200 plus completed held-out 40

Table S5: Cumulative nested-contract survival (percent). Cross-model AlgoTutorGen rows use final-quality artifacts with recorded retries. The main paired comparison uses one frozen selected-final page per task and is not claimed to be call- or token-matched to Direct HTML.

Method	C1	C2	C3	C4	C5	C6
AlgoTutorGen main	100.0	100.0	100.0	99.0	99.0	99.0
Direct HTML main	100.0	94.0	74.5	54.0	49.0	49.0
AlgoTutorGen Flash final	100.0	100.0	100.0	100.0	100.0	100.0
Direct Flash	99.0	95.0	81.0	65.5	59.0	59.0
AlgoTutorGen GLM final	99.0	99.0	99.0	98.0	98.0	98.0
Direct GLM	100.0	91.0	53.0	38.0	17.5	17.5
AlgoTutorGen Kimi final	97.5	97.5	97.5	97.0	97.0	97.0
Direct Kimi	99.5	95.0	68.0	52.5	43.5	43.5

Table S6: Trace–scene–runtime projection audit.

Set	Artifact pass	Equivalent frames
Main	200/200	9,421/9,421
Completed held-out	40/40	4,568/4,568
Long-trace	54/54	41,119/41,119
Total	294/294	55,108/55,108

Twenty stratified artifacts were recompiled and rerendered ten times each. Every artifact produced one canonical projection hash and one render hash across reruns. These observations establish representation-level agreement and tested-environment stability; they do not independently establish that every source-trace event is the intended algorithm step or that pixels are formally verified.

Semantic Mutation and Overlay Isolation

The mutation suite distinguishes changes that must be rejected from transformations that should be accepted. Ambiguous event deletions are not automatically classified as semantic faults because a deleted explanation or repeated mark may be redundant. Reordered unordered-set results are compared with a case-aware equivalence relation.

Table S7: Contract discrimination on the defined mutation suite.

Class	Expected	Observed
Semantic violations	Reject	2,198/2,198
Teaching-text rewrites	Accept	195/195
Visual-metadata changes	Accept	195/195
Equivalent reorderings	Accept	2/2
Preserving total	Accept	392/392

Table S8: Teaching-overlay isolation. Outcomes preserve the canonical algorithm-state hash unless marked as rejected or warned.

Condition	Outcome
Four legal overlay variants	372/372 each
Illegal `final_answer`/`state` fields	372/372 sanitized
Negative step	372/372 rejected
Nonexistent step	372/372 warned
Cross-model GLM overlay	369/369

The legal variants are original, concise, detailed, and random text overlays; their text changes do not redefine algorithm state. The existing sanitizer removes the tested protected writes, the schema rejects negative indices, and step-not-found warnings prevent unbound teaching steps. For the GLM overlay, 169 cases apply fully and 200 apply partially because model-specific traces expose different step sets. The illegal-field result is specifically sanitization, not a claim that every unknown field is rejected by schema-level `extra=forbid`. Complete discrimination is claimed only for these defined finite suites.

Pedagogical Noninterference Stress Test

Table S9: Browser noninterference stress-test coverage.

Quantity	Count
Unique pages	240
Random action sequences	24,000
Total actions	1,561,298
Pure pedagogical actions	435,859
Navigation/variant actions	1,125,439
Pages passed	240/240
Overlay artifacts passed	240/240
Observed violations	0

Pure actions are correct submission, wrong submission, hint, show answer, and clear learning log. Navigation actions are next, previous, timeline selection, reset, and variant selection. Each page receives 100 random sequences of 30–100 actions. Pure actions must preserve artifact, state, and step hashes; navigation must preserve the artifact and select the expected verified frame. The result is that no counterexample was found, not that all possible browser executions are proved noninterfering.

Local Resume Versus Global Restart

Both policies use the same structured output space, validators, and maximum of three policy decisions. Local keeps the current solution specification, repairs it, then reruns materialization and teaching. Global discards the specification and independently regenerates it before rerunning the same downstream work.

Table S10: Recovery outcomes, paired success counts, and token decomposition. Calls and tokens are per successful page and include final-failure costs; time is time-to-valid on successful cases.

A. End-to-end outcomes

Model	Policy	Succ.	Calls	Tokens	Time	p
Flash	Local Resume	38/50	6.63	71,369	172.9 s	0.4545
Flash	Global Restart	42/50	5.50	62,256	194.2 s	
GLM	Local Resume	42/50	6.69	92,385	533.8 s	1.0000
GLM	Global Restart	43/50	6.65	96,186	558.2 s	

B. Paired success counts

Model	Both	Local only	Global only	Neither
Flash	32	6	10	2
GLM	37	5	6	2

C. Token decomposition per successful page

Model	Policy	Spec/repair	Teaching	Total
Flash	Local Resume	25,031	46,338	71,369
Flash	Global Restart	25,781	36,476	62,256
GLM	Local Resume	37,342	55,043	92,385
GLM	Global Restart	47,462	48,724	96,186

Flash favors Global numerically in success and tokens per success. GLM gives Local slightly lower token and time cost but one fewer success. Neither paired success difference is significant. The negative result does not contradict the ideal checkpoint theorem: the implementation violates its no-recomputation premise because repair rematerializes the specification and reinvokes teaching.

Additional Reliability Evidence

Held-Out Tasks and Full-Sample Replay

Table S11: Additional task/input robustness.

Set	AlgoTutorGen	Direct
Held-out generation	39/40	40/40
Held-out Machine OK	39/40	18/40
Held-out self-contained	40/40	40/40
Frozen tracker replay	626/646	–

The held-out set contains 40 new cases from 15 families. The Machine OK difference is +52.5 percentage points (95% paired-bootstrap CI [37.5, 67.5], exact McNemar $p = 9.54 \times 10^{-7}$). The targeted retry used to complete the semantic audit does not retroactively change 39/40 generation. The 646-sample replay reuses frozen solvers/trackers across all stored inputs and has 20 failures; it is an unbalanced robustness set, not 646 independent main-table tasks.

Non-Degenerate Ablations and Browser Repair

Table S12: Non-degenerate 50-task ablations. Both outputs are self-contained under the corrected resource parser.

Condition	Full	Ablation
Direct-to-SceneGraph	49/50	1/50
VerifiedTrace-to-LLM-HTML	49/50	0/50

Direct-to-SceneGraph isolates the fixed runtime without an executable trace and validation; its 96-point difference has a 95% CI of [90, 100] and exact McNemar $p = 7.11 \times 10^{-15}$. VerifiedTrace-to-LLM-HTML isolates unconstrained page generation after a verified trace: its visible answer is 50/50, but interaction reachability and Machine OK are 0/50. On these 50 tasks, neither a fixed renderer nor a verified trace alone reproduces the full-chain reliability.

Table S13: Direct whole-HTML browser-repair budget.

Calls	Machine OK	Avg. tokens	Avg. time
1	106/200	19.7k	207.2 s
2	10/200	36.8k	347.2 s
3	15/200	53.7k	477.6 s
5	6/200	87.2k	733.9 s

All 1,000 repair attempts are self-contained after correcting the resource parser. More calls do not monotonically improve this evaluated whole-page rewrite policy; the result does not imply that every browser-repair method fails.

Qualitative Results and Case Studies

The figures in this section use a separate 23-case English qualitative package with five frozen methods per case. It is not substituted for the Full-200 main experiment. Figure S1 follows one verified dynamic-programming artifact from task contract to terminal result; Figure S2 aligns four methods on the same graph task; and Figure S3 samples four algorithm families. Every colored check label is copied from that artifact's nine-field `audit.json`; screenshots document visible layout only and are not treated as proof of hidden interaction behavior.

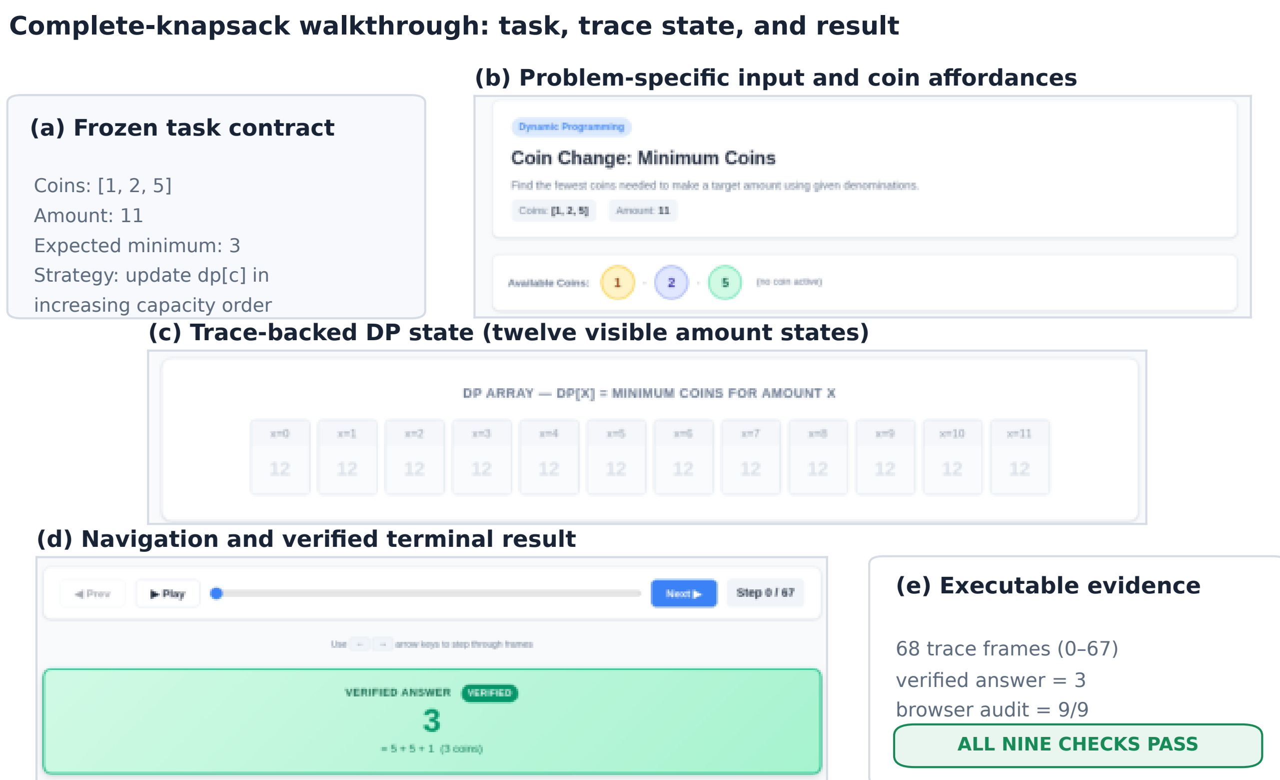

Figure S1: Executable walkthrough for complete-knapsack coin change. The concrete input is coins $[1, 2, 5]$ and amount 11; the expected minimum is three coins. The selected ALGOTUTORGEN page exposes the task-specific coin view, a 12-state DP array, navigation over 68 trace-backed frames (indexed 0–67), and the verified answer 3. Its qualitative-package artifact passes all nine recorded browser checks. The figure crops one frozen screenshot into readable regions; it does not reconstruct or edit the underlying algorithm state.

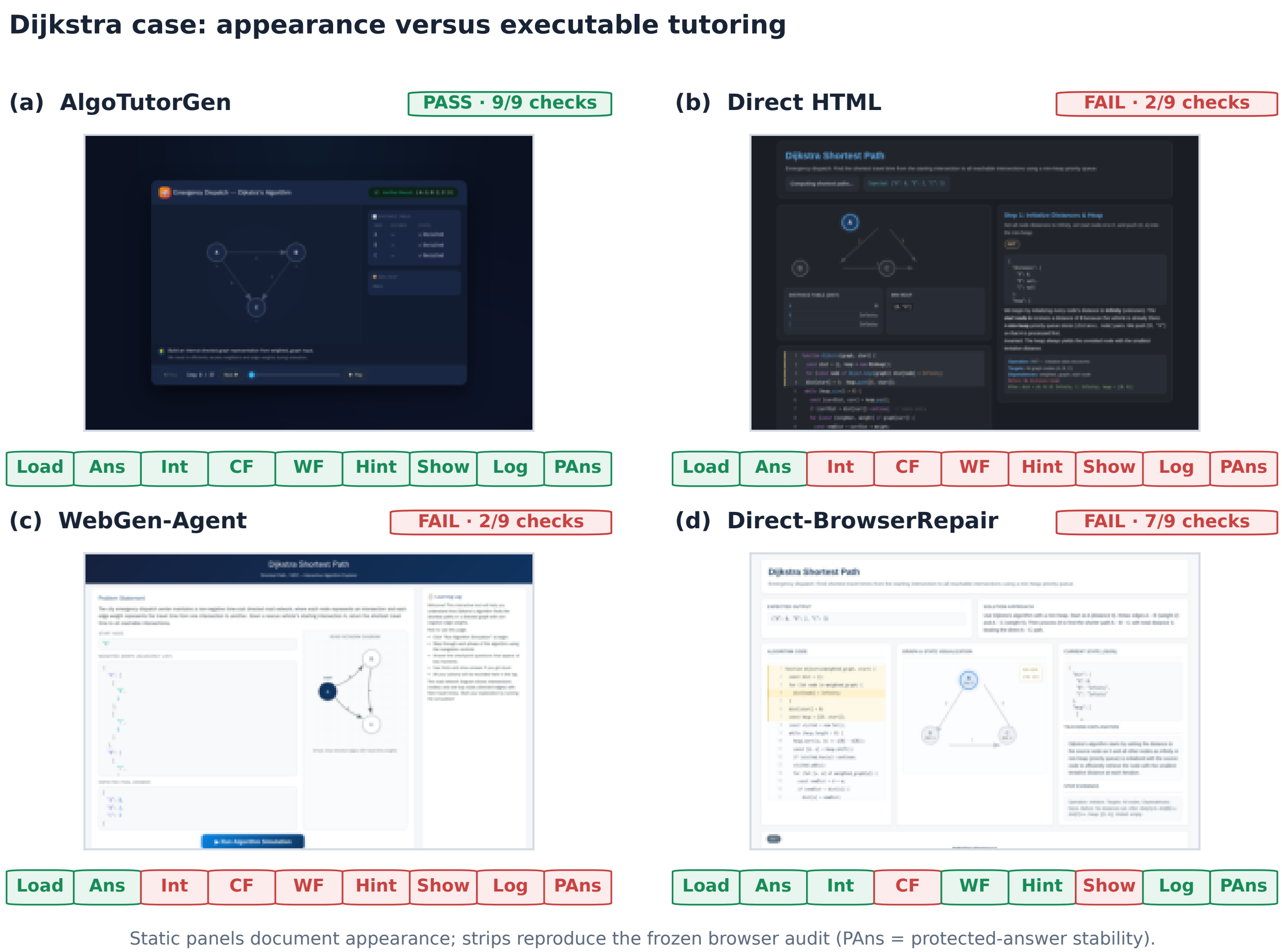

Figure S2: Aligned qualitative comparison on one Dijkstra task. All four pages receive the same directed graph, start node, and expected distances. AlgoTutorGen passes 9/9 recorded checks. Direct HTML and WebGen-Agent each pass only load and visible-answer checks; their interaction, both feedback directions, hint, answer reveal, log, and protected-answer checks fail. The one-call Direct-BrowserRepair page passes 7/9 and fails correct feedback and answer reveal. The static panels show that visually plausible output is not equivalent to executable tutoring reliability; the colored strips report browser outcomes rather than pixel judgments. HTMLCure is omitted because its Dijkstra HTML and screenshot are byte-identical to the Direct HTML output.

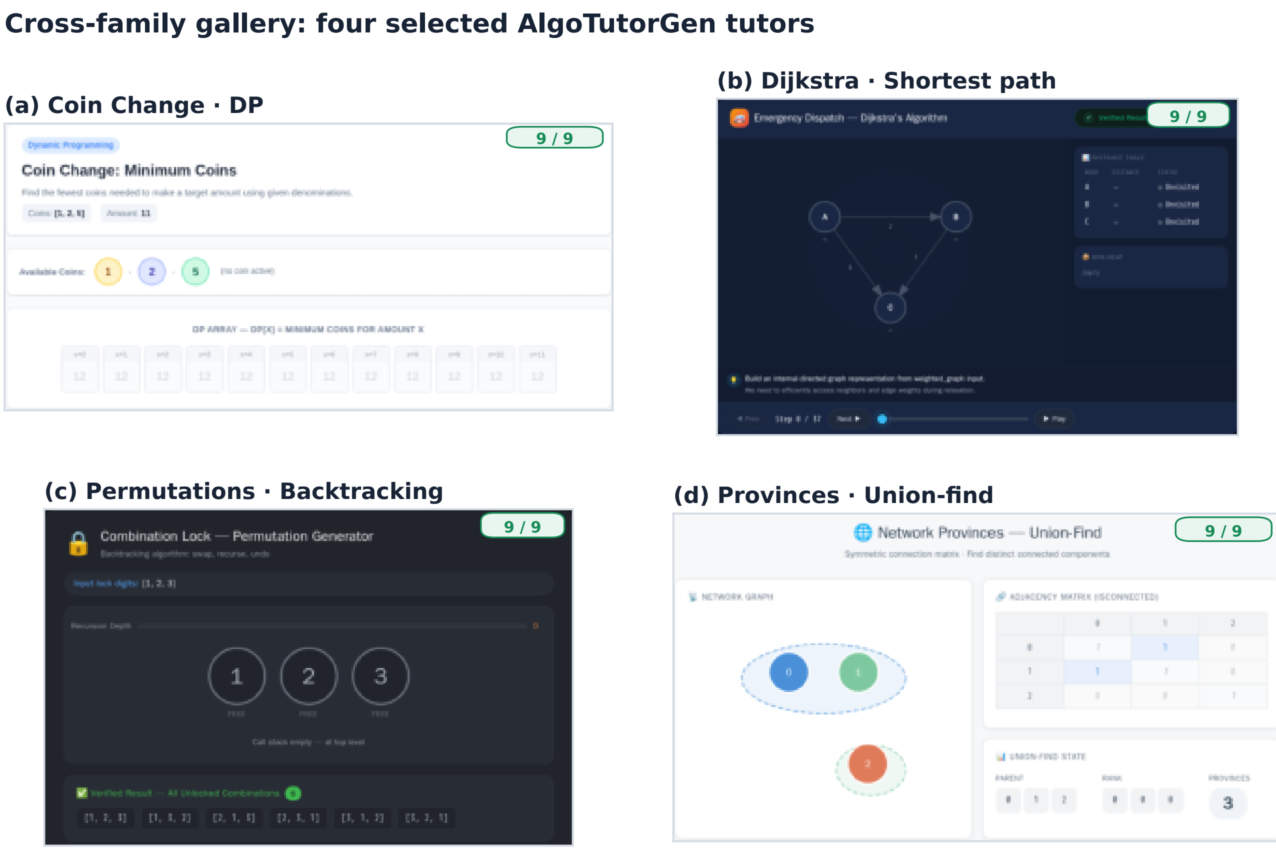

Figure S3: Cross-family gallery of selected AlgoTutorGen artifacts from dynamic programming, shortest paths, backtracking, and union-find. Each frozen page passes the qualitative package's nine browser checks. The views use different problem-specific structures—coin/DP states, a weighted road graph with distances and heap, a permutation search state, and connected components with union-find state—rather than a single repeated visualization template.

Original Main-Run Generation Cost

Table S14: Original DeepSeek-V4-Pro generation boundary. No stable price configuration was frozen, so monetary cost is not reported.

Condition	Calls	Tokens	Tokens/task
Selected-final	1,066	15,369,433	76.8k
All attempts	1,151	16,870,557	–
Direct HTML	222	4,385,641	21.9k

The extra compute is part of the evaluated system. The fixed runtime and compiler also embody engineering unavailable to the Direct generator; ablations reduce but do not remove this implementation asymmetry.

Long-Trace Scalability

All 54 artifacts materialize, while 52 complete browser measurement. The exporter currently embeds complete frames in a self-contained document, so repeated state dominates size and load cost.

Table S15: Long-trace browser costs; all entries are means.

Scale	Frames	HTML	Load	Step	Heap
Small	104.4	1.40 MB	120 ms	14.7 ms	6.3 MB
Medium	543.3	23.25 MB	1,933 ms	45.4 ms	59.1 MB
Large	1,636.7	160.42 MB	8,354 ms	101.3 ms	185.6 MB

The large KMP page contains 3,063 frames and 581 MB of HTML. The large sliding-window-unique page contains 5,533 frames and 1.08 GB of HTML. Both exceed the 60-second load budget. These results motivate event deltas, frame virtualization, lazy loading, or state compression when scaling beyond the evaluated regime.

Failure Analysis

Selected-Final Browser Failures

The main AlgoTutorGen condition generates valid artifacts for all 200 tasks, but two pages fail the final conjunction. `stack_valid_parentheses_full_core` fails wrong-answer feedback; `string_longest_common_prefix_full_core` fails both correct- and wrong-answer feedback. Both pages pass load, visible answer, reachable interaction, hint, answer reveal, learning log, and protected-answer stability. The first failing boundary is therefore the teaching-feedback contract, not solver materialization or scene compilation.

The final cross-model failures separate missing artifacts from generated pages with incomplete feedback. Flash has no final failure. GLM fails to generate valid artifacts for two tasks and has two generated pages with bidirectional-feedback failure. Kimi fails to generate five artifacts and has one generated page with bidirectional-feedback failure. The held-out failure is a materialization/generation failure; its missing page consequently fails all downstream browser checks. These categories are not merged into one generic "page failure."

Cross-Input Replay

Frozen sample-0 solvers and trackers pass 626/646 stored inputs. Table S16 groups the 20 failures by the first structural boundary. The full task/sample list is retained in the detailed result ledger.

Table S16: Cross-input replay failure taxonomy. Counts are mutually exclusive.

First structural boundary	Count	Representative mechanism
Algorithm/edge-input implementation	12	Empty arrays or trees, long pattern, duplicate-word counting, unsupported DSL call
Equivalence normalization	2	Different valid topological order or permutation-set order
Schema or target reference	4	Missing slice target; tree node/ID assumptions
Unreachable-value representation	2	Extra or inconsistent infinity entries for disconnected nodes
Total	20	426/446 additional inputs pass

The replay result is a direct limitation of instance-conditioned generation: passing one concrete input does not prove that the frozen code generalizes to every edge input of the same task. Some failures require stronger family-aware equivalence and schema normalization; others require fixing the generated algorithm or tracker itself.

Baseline Failure Mechanisms

Direct HTML has 12 load failures caused by JavaScript syntax, duplicate declarations, invalid escapes, or null-DOM operations. Among accepted HTMLCure rewrites, 125 introduce a Google Fonts request and therefore fail the strict self-contained condition when external requests are blocked. WebGen-Agent has one rendering timeout. These are protocol-specific mechanisms, not evidence that the underlying systems always fail. The Dijkstra case in Figure S2 was selected because every page loads and shows the answer, localizing the contrast to interaction and teaching behavior rather than a trivial parse failure.

Secondary Visual Evaluation

Table S17: Stage2 versus Direct under the same machine-rated rubric. Only instructional design is significant after Holm correction.

Dimension	Stage2	Direct	Holm p
Overall	4.611	4.596	–
Problem alignment	4.835	4.825	0.856
State readability	4.385	4.505	0.059
Process clarity	4.320	4.400	0.326
Instructional design	4.905	4.655	1.56×10^{-5}

The result does not support visual superiority on every dimension. Direct is numerically higher on state readability and process-transition clarity. The evaluator is one vision-language model, so these results remain secondary until human calibration is completed.

Reproducibility Boundaries

The repository snapshot accompanying this draft contains the implementation, benchmark definitions, prompt sources, detailed result ledger, and a selected English qualitative package. The most useful included entry points are:

- `docs/EXPERIMENT_RESULTS_DETAILED.md`, which records the final reporting hierarchy, configurations, totals, and source-report paths;
- `algolab/generation/prompts/` and the dynamic prompt builders in `algolab/generation/`;
- `artifacts/method_comparison_samples_en/manifest.json`, covering 23 cases, five methods, 115 screenshots, and their audit records; and
- `artifacts/paper_case_studies_en/figure_manifest.json` plus `artifacts/paper_case_studies_en/PROMPT_TEMPLATES_EN.md`, which freeze the qualitative-figure hashes and English prompt renderings.

The large raw experiment directories referenced by the detailed ledger are not duplicated in this lightweight source snapshot. Therefore this snapshot alone reproduces the code path, released qualitative artifacts, and paper figures, but not every aggregate number from raw logs. A complete archival release must include the referenced JSON reports and their SHA-256 manifest; until then, the aggregate tables should be interpreted as results from the frozen evaluation archive rather than independently rerunnable from this checkout alone.

Browser evaluation uses a 1365×900 viewport, waits 500 ms after DOM content load, searches at most 80 timeline steps for a checkpoint, and blocks external HTTP requests in strict conditions. Paired bootstrap uses seed 20260713 with 10,000 draws; noninterference sequences use seed 20260714. The recorded browser build is Playwright Chromium 1223. Remote model versions can drift, and self-contained artifacts cannot reproduce a retired remote service exactly. Selected-final reporting preserves retry provenance rather than treating every page as a first-pass generation.

Prepared but Unlabeled Human Protocols

No protocol below has completed labels or participants:

- **Machine-evaluator calibration:** 30 tasks $\times$ 4 methods, or 120 blind pages, prepared for two human annotators; no precision or recall result exists.
- **Independent trace audit:** 40 tasks across 23 families prepared for two reviewers; no critical-error rate exists.
- **Expert comparison:** 3 experts $\times$ 30 pairs planned; there is no preference or usability result.
- **Student study:** 24 students $\times$ 12 trials planned, subject to any required ethics process; there is no learning outcome.

These protocols are future evaluation infrastructure. No missing human labels are imputed, and no learning, retention, transfer, expert-preference, or human-calibrated trace-correctness claim is made.